DEEP LEARNING ARCHITECTURE SERIES

Gated Recurrent Units (GRU)

1. OVERVIEW & CORE CONCEPT

Introduced by Cho et al. in 2014, the **Gated Recurrent Unit (GRU)** is a gating mechanism for Recurrent Neural Networks (RNNs). Designed as a streamlined alternative to Long Short-Term Memory (LSTM) networks, GRUs address the vanishing gradient problem encountered when processing sequential data and learning long-range dependencies.

Unlike traditional LSTMs which maintain separate cell and hidden states, the GRU merges these into a single hidden state vector. It manages memory using two internal gating mechanisms: the **Update Gate** and the **Reset Gate**.

2. ARCHITECTURE & SIGNAL FLOW



x_t Reset Gate $r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$ Update Gate $z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$ Candidate State $\tilde{h}_t = \tanh(W \cdot [r_t \odot h_{t-1}, x_t])$ State Output h_t

3. KEY GATING COMPONENTS

Reset Gate (r_t)

Determines how much past information to forget. When $r_t \approx 0$, the model drops the previous hidden state, allowing it to discard irrelevant temporal details.

Update Gate (z_t)

Controls the balance between the previous hidden state and the newly computed candidate state. It performs the dual role of LSTM's forget and input gates.

Candidate Hidden State ($\tilde{h}_t$)

Calculates current context using input x_t and past memory scaled by the Reset Gate, using a non-linear `tanh` activation function.

Final Hidden State (h_t)

Combines past context h_{t-1} and new candidate state $\tilde{h}_t$ via linear interpolation governed directly by update gate z_t.

4. MATHEMATICAL FORMULATION

At time step **t**, given input vector $\mathbf{x_t}$ and previous state $\mathbf{h_{t-1}}$, the internal computations follow:

Update Gate: $z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$
Reset Gate: $r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$
Candidate Hidden State: $\tilde{h}_t = \tanh(W_h \cdot [r_t \circledcirc h_{t-1}, x_t] + b_h)$
Hidden State Output: $h_t = (1 - z_t) \circledcirc h_{t-1} + z_t \circledcirc \tilde{h}_t$

* Note: σ denotes the sigmoid activation function; ⊙ represents Hadamard (element-wise) multiplication.

5. COMPARISON: GRU VS. LSTM

Feature	Gated Recurrent Unit (GRU)	Long Short-Term Memory (LSTM)
Gates	2 (Reset, Update)	3 (Input, Forget, Output)
State Vectors	Single Hidden State (h_t)	Cell State (c_t) & Hidden State (h_t)
Parameters	~25% fewer parameters	Higher parameter count
Training Speed	Faster convergence per epoch	Slower execution, larger memory footprint
Data Efficiency	Performs better on smaller datasets	Superior expressive power on massive datasets

6. ADVANTAGES & USE CASES

By reducing parameter overhead, GRU models provide computationally efficient performance without significantly sacrificing sequential modeling capacity compared to conventional LSTMs.

Primary Practical Applications:

- **Natural Language Processing:** Machine translation, sentiment analysis, and text classification.
- **Time Series Forecasting:** Financial marker modeling, weather predictions, and sensor data processing.
- **Speech Recognition:** Acoustic modeling and real-time audio sequence handling.
- **Resource-Constrained Systems:** On-device edge ML where memory footprint and execution latency are critical.